

IMPACT OF AI-POWERED POLICING ON RACIAL BIAS

Fareeza Ali, Ishika Gobin, Jackson Howes, Dana Kim
Mentor: Dr. Purna Gamage

Background

Predictive policing systems are increasingly used to guide law enforcement decisions. The NYPD implemented *Patternizr* in 2013, a proprietary algorithm designed to identify crime patterns. However, concerns exist that such systems may reinforce historical racial biases embedded in policing data. This study evaluates whether *Patternizr* altered racial disparities in policing outcomes across the NYC precincts.

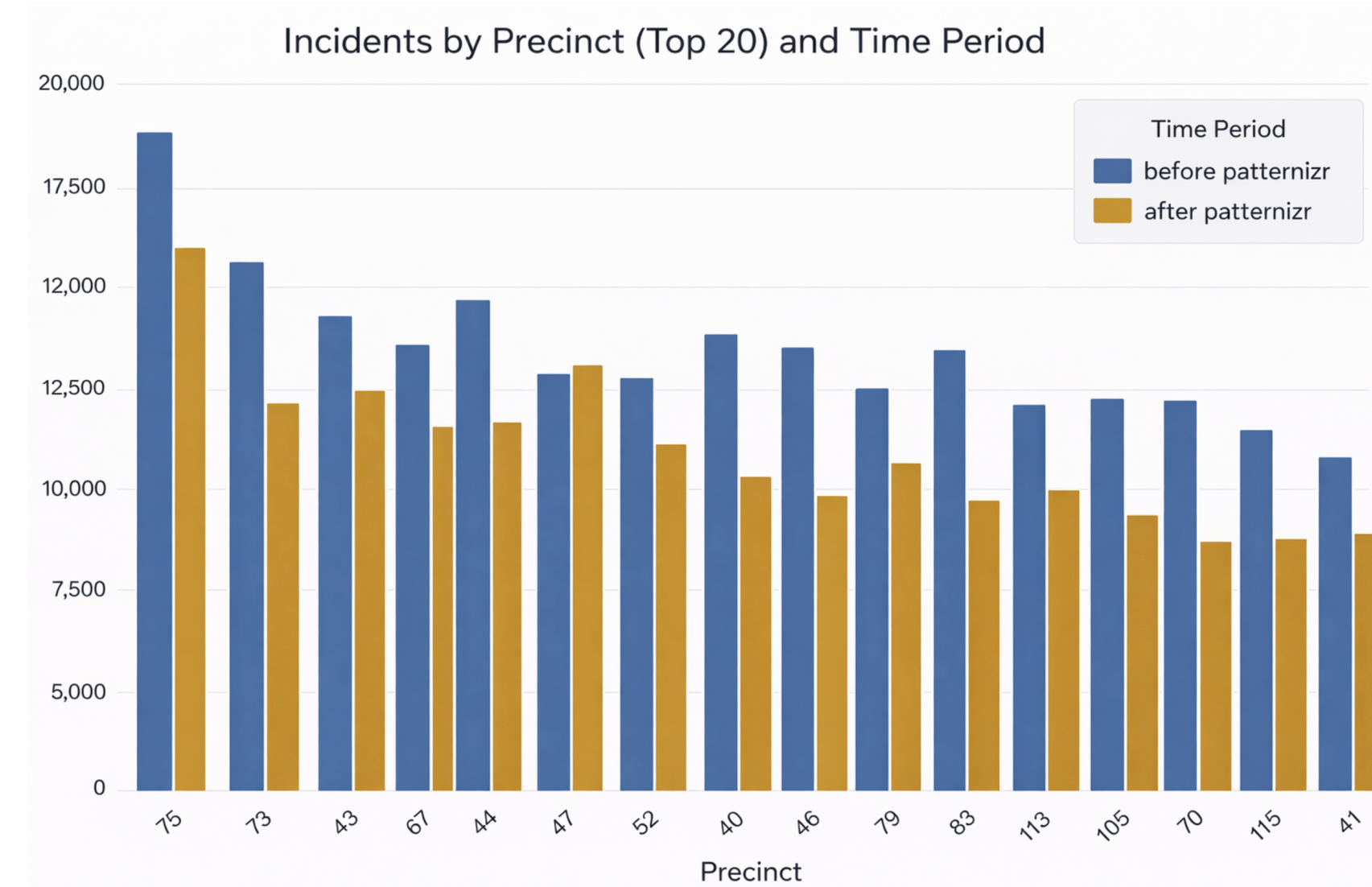


Fig. 1. Top 20 Precincts with the most complaints before and after Patternizr

Methods

- Bootstrapping for missingness
- Statistical Tests
 - T-test
 - Chi-Square test + post-hoc of residuals
 - Two-way ART ANOVA + post-hoc of pairwise contrasts
- Time series forecasting: ARIMA/SARIMA
- Key variables
 - Crime occurrence date
 - Crime reported date
 - Police precinct
 - Suspect race

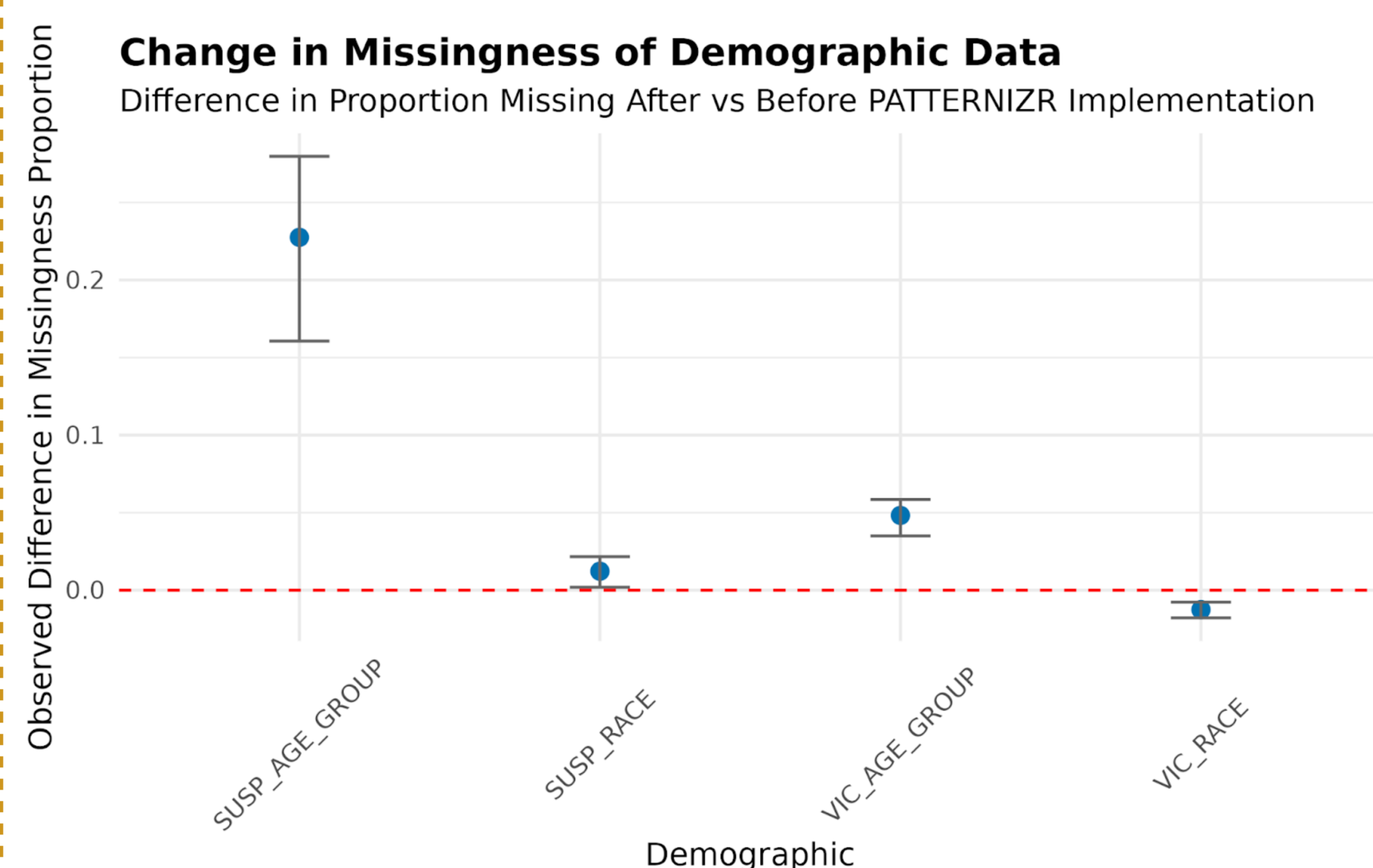


Fig. 2. Bootstrapped Change in Proportion of Missingness after Patternizr

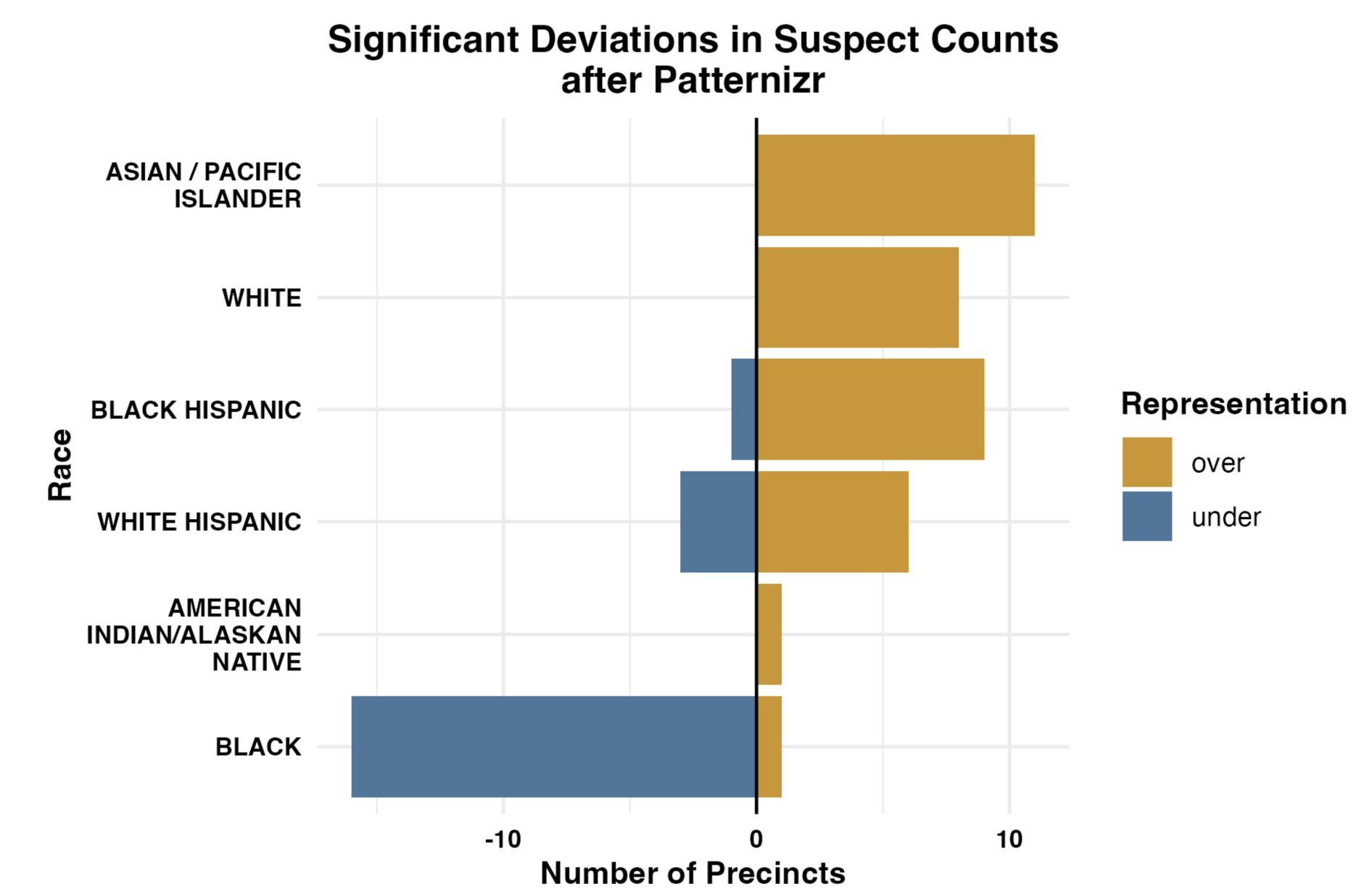


Fig. 3. Count and Direction of Statistically Significant Racial Group Proportions After Patternizr across 20 Precincts

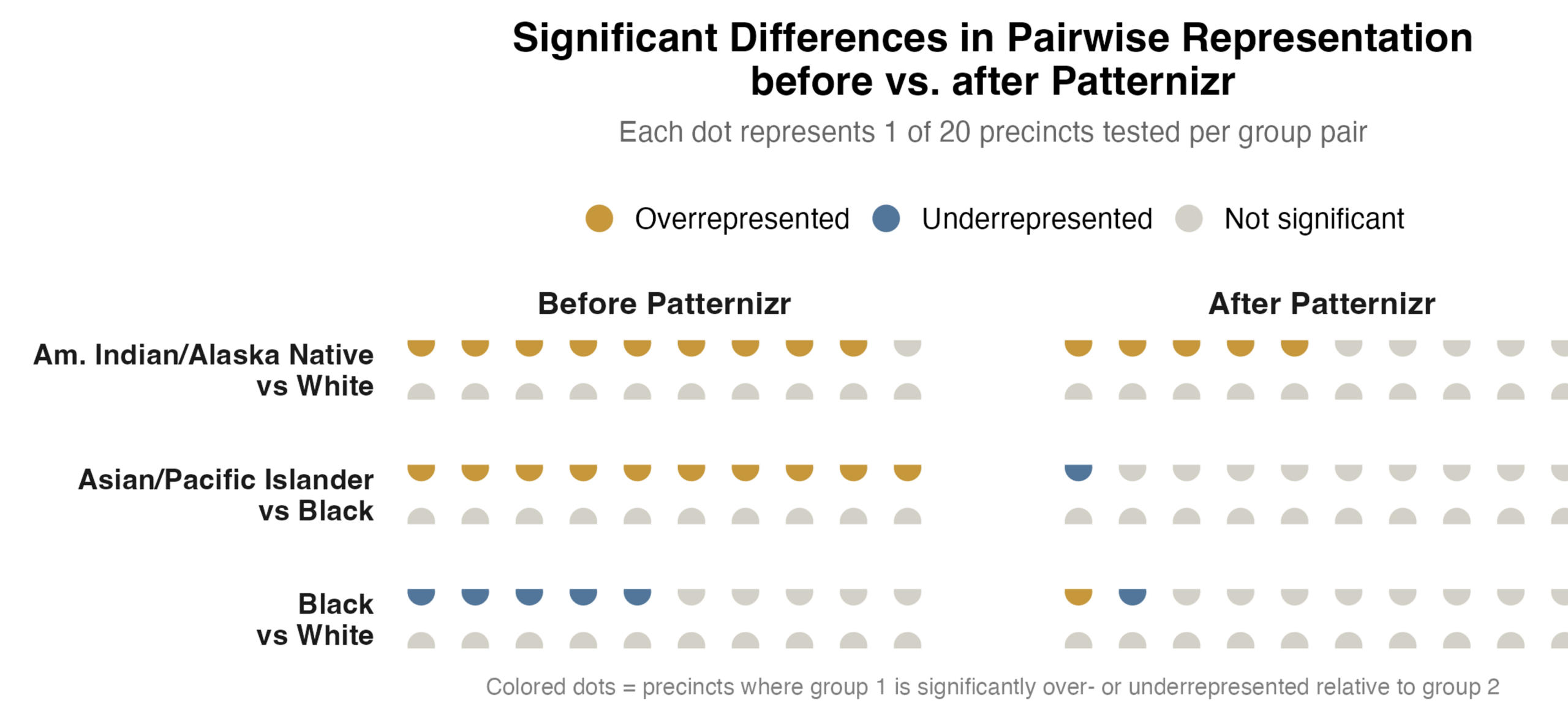


Fig. 4. Count and Direction of Statistically Significant Racial Pairings for Top 3 Most Commonly Found Pairings

Key Findings

Chi-Square

The implementation of *Patternizr* was associated with changes in the NYPD suspects' racial composition, meaning certain racial groups experienced more interactions with police. In these shifts in racial composition, it was most often found that historically marginalized racial groups became underrepresented and historically privileged groups became overrepresented.

ANOVA

The impact of *Patternizr* on the monthly average number of complaints also differed depending on the race of the suspects. Among the racial pairings that most frequently showed these disproportionate number of complaints, historically marginalized groups tended to become overrepresented relative to their counterparts (in contrast to the Chi-Square results).

Time Series 1:

Projected reporting delays stabilized around 15 days for Black suspects versus 35–40 days for White suspects – meaning crimes involving Black suspects were reported roughly 3x faster. Neither series shows a meaningful shift after *Patternizr's* implementation. Wide prediction intervals in both models reflect high volatility, likely because the data does not differentiate between crime types.

Time Series 2:

Total projected complaints involving Black suspects were 5–7x higher per month compared to White suspects. Both groups show a sharp, sustained increase around 2014–2015, coinciding with *Patternizr's* widespread adoption: approximately +700/month for Black suspects and +100/month for White. While the proportional increase was relatively equitable, the absolute gap means NYPD policing activity continues to disproportionately involve Black suspects.

Forecast of Average Reporting Delay by Race

Black: ARIMA(2,0,1)
White: ARIMA(1,0,1)

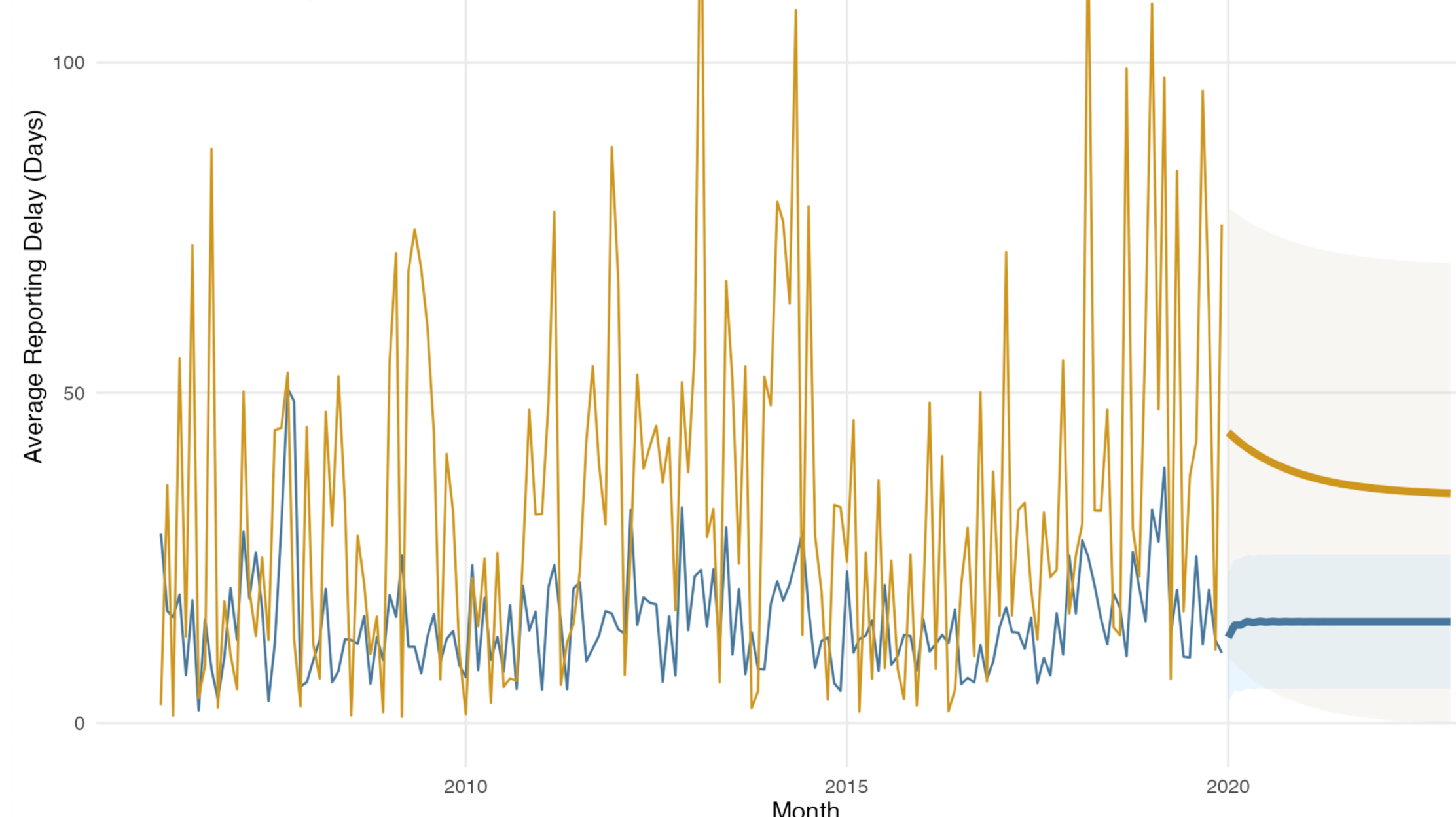


Fig. 5: Best fit ARIMA time-series model forecasts of crime report delays of incidents with Black suspects vs. White suspects.

Forecast of Total Monthly Complaints by Race

Black: SARIMA(1,0,0)(2,1,2)[12]
White: SARIMA(0,1,1)(0,0,2)[12]

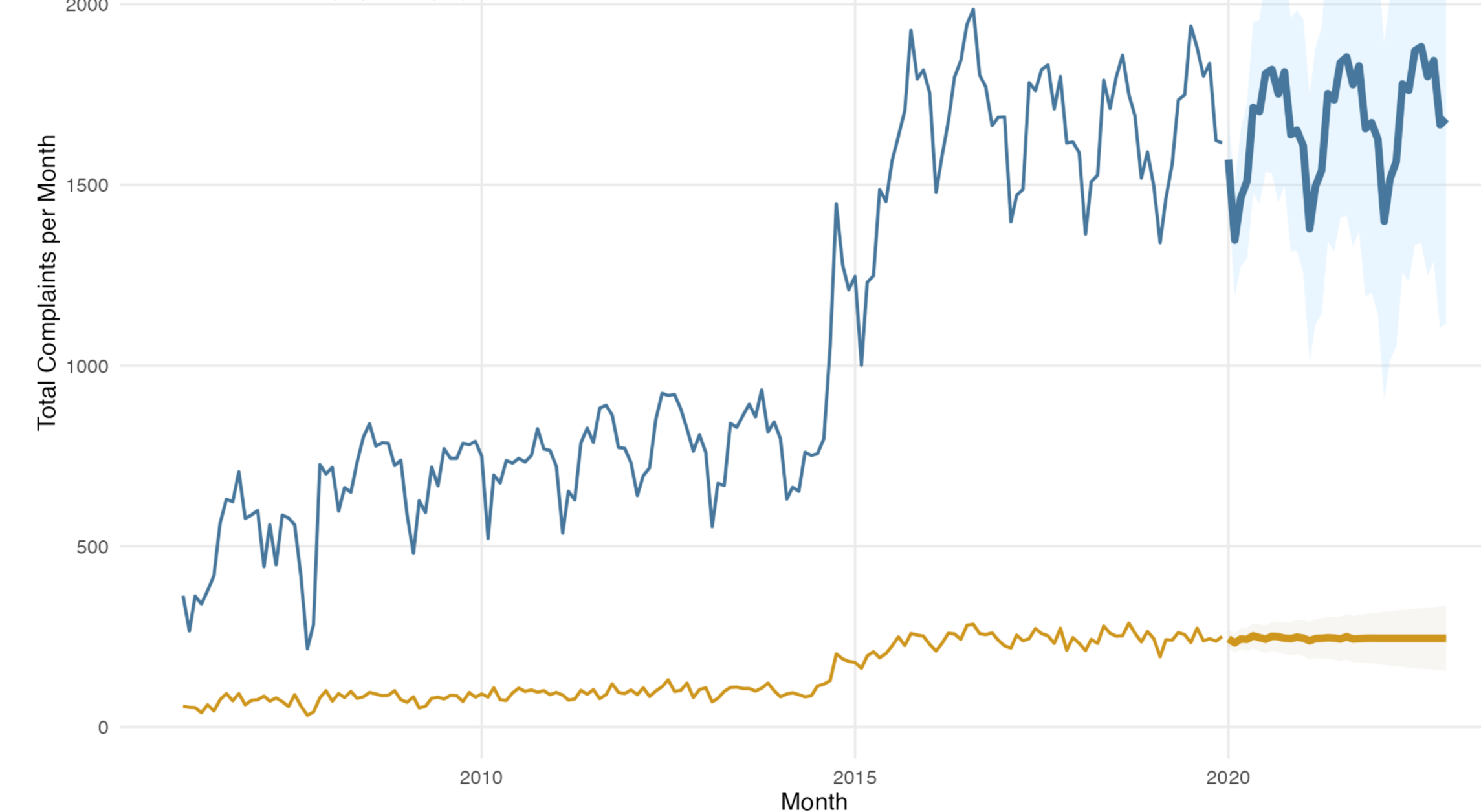


Fig. 6: Best fit Seasonal ARIMA time-series model forecasts of crime complaints involving Black suspects vs. White suspects

Discussion

Although the Chi-Square results in isolation seem to indicate that *Patternizr* “corrected” bias, it is more likely that this “correction” is on an already extremely biased overrepresentation of Black suspects before *Patternizr*. This reasoning is supported by both the contrasting results of our ANOVA and the wealth of research on disproportionate policing of marginalized racial groups.

Limitations

- No downstream outcomes**
We do not examine arrests, convictions, or other downstream justice outcomes.
- No breakdown of crime-type**
Results are not disaggregated by felony type, which may influence observed racial differences.
- Unexplained reporting delay**
Differences in reporting delays may reflect external social factors not captured in the data.

Future work: disaggregate by felony crime type to assess whether findings persist at that level

Conclusion

- Predictive policing systems like *Patternizr* do not eliminate racial disparities and may reinforce them.
- Evidence suggests relative exacerbation of disparities for marginalized groups.
- Highlights the need for:
 - Algorithmic transparency to the public
 - Careful evaluation of AI-powered policing systems before and after deployment

Predictive systems in policing must be critically assessed to avoid amplifying historical inequities

References and Full Paper



<https://bit.ly/Patternizr5100>